School of Computing and Information System

Project: Where is AI needed most in Programming Education?

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2024	H1 - 2025	H2 - 2025	2026
Completed MC-IT AMCIS 2024 Top 25% Complete Papers	Confirmed as PhD Candidate Conducted preliminary research on Socratic AI Tutor	Obtained result from first study Industry Internship	Conducting current research on AI importance with mixed method approach

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Overview

1 Motivation for Initial Study

2 Platform walkthrough

3 Initial result

4 The need for the current study

5 Methodology

6 Preliminary Results

7 Planned Works

Motivation for Socratic AI Tutor Study

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The Bad Side of AI in Education

- Over-reliance on technology reduces knowledge gain
- The issue of “Cognitive Debt” (Kosmyna et al., 2025)
- Hallucination, Over-confidence in incorrect responses, bias/misinformation propagation

Kosmyna, N., Hauptmann, E., Yuan, Y. T., Situ, J., Liao, X., Beresnitzky, A. V., Braunstein, I., & Maes, P. (2025, June 10). Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task. arXiv.org. https://arxiv.org/abs/2506.08872

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The Potential Silver Bullet

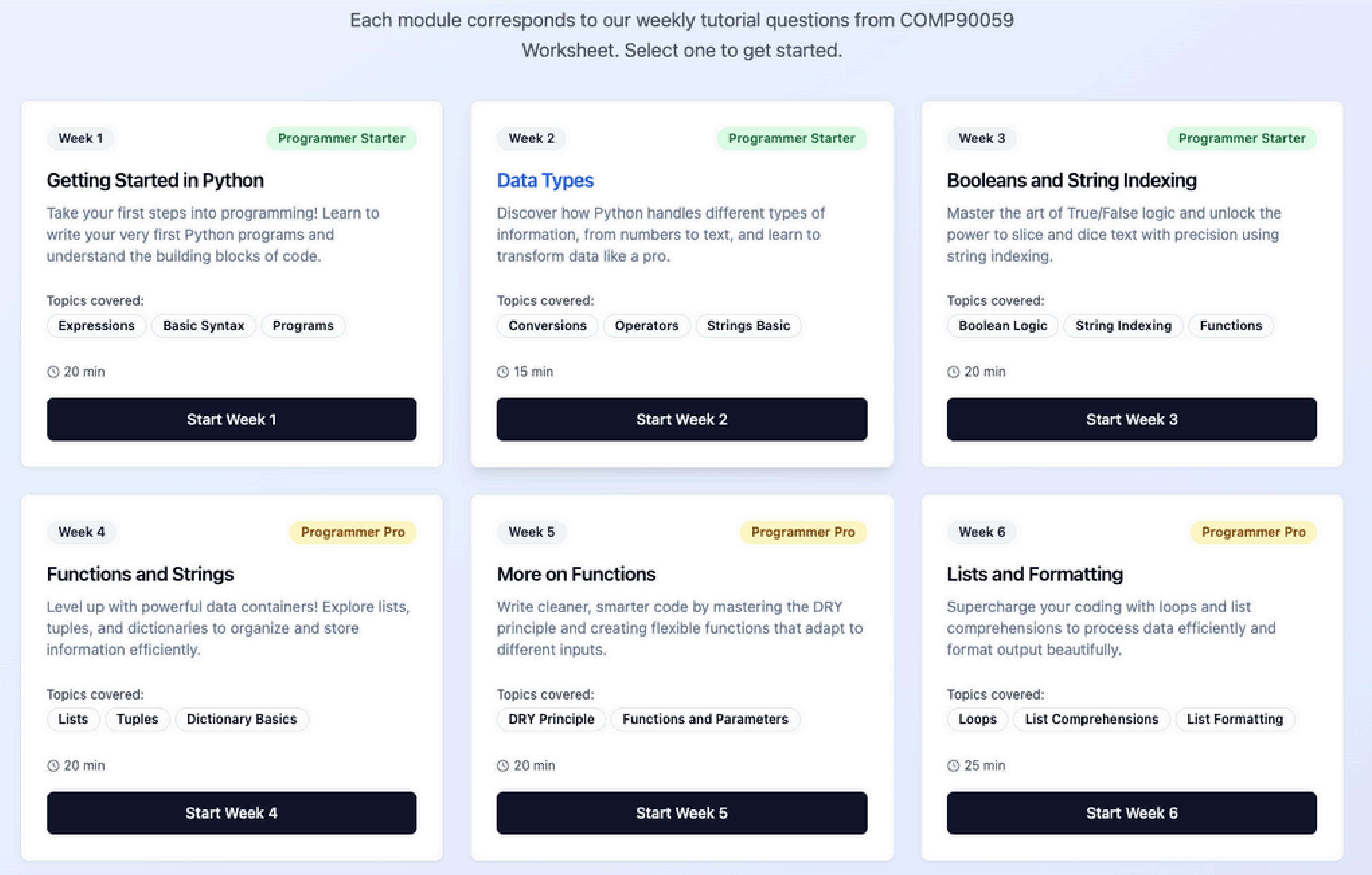
- Have AI become an active tutor, challenge students and not providing answers
- The “Socratic Method” implemented via AI instructions

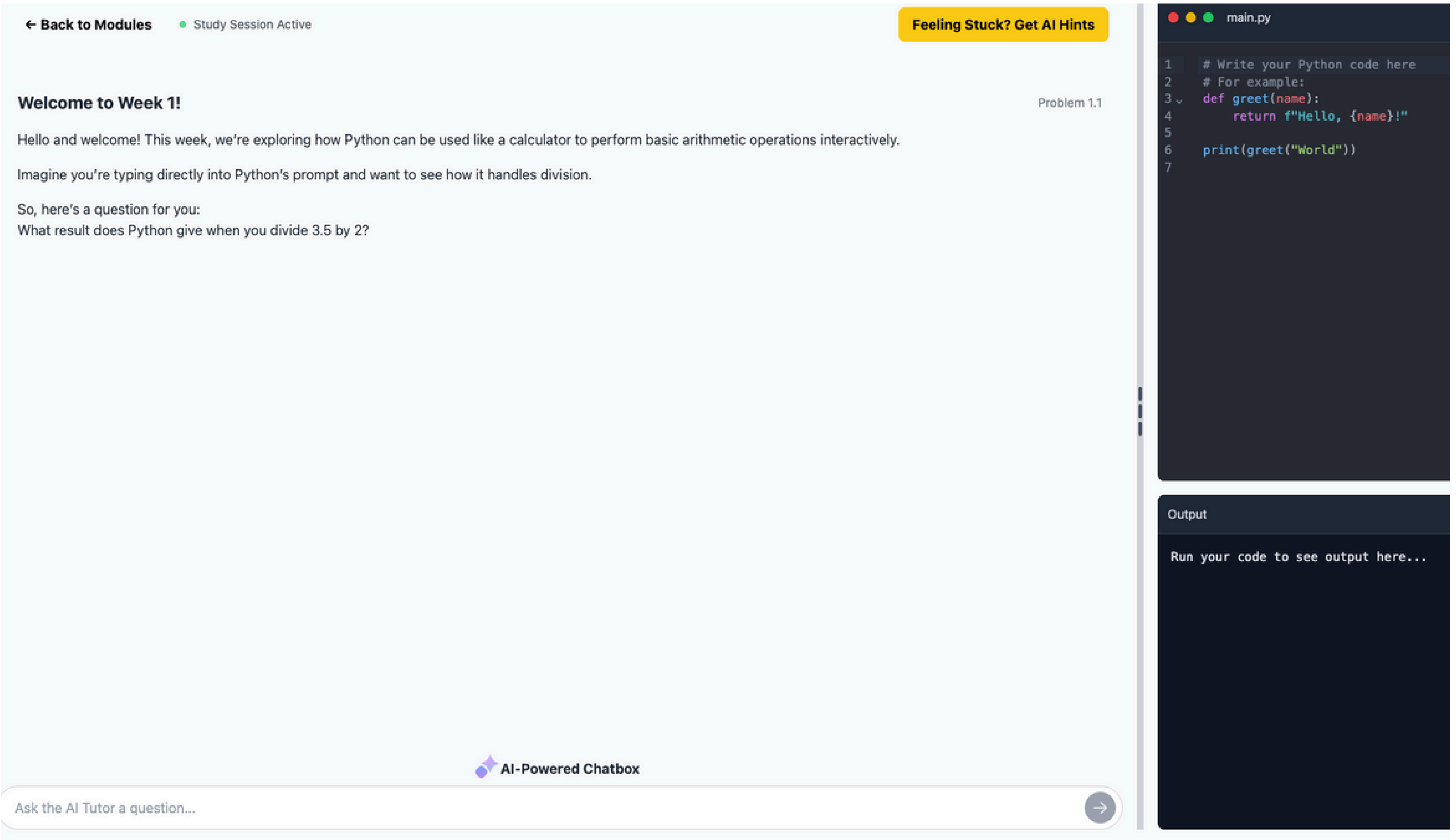
  More recently: Khanmigo, ChatGPT Learning Mode

- The desired effects remain hypothetical

Degen, P., & Asanov, I. (2025, August 7). Beyond automation: socratic AI, epistemic agency, and the implications of the emergence of orchestrated Multi-Agent Learning architectures. arXiv.org. https://arxiv.org/abs/2508.05116
Duelen, A., Jennes, I., & Van Den Broeck, W. (2024). Socratic AI against Disinformation: Improving critical thinking to recognize disinformation using Socratic AI. ACM International Conference on Interactive Media Experiences, 375–381. https://doi.org/10.1145/3639701.3663640
Fakour, H., & Imani, M. (2025). Socratic wisdom in the age of AI: a comparative study of ChatGPT and human tutors in enhancing critical thinking skills. Frontiers in Education, 10. https://doi.org/10.3389/feduc.2025.1528603
Seo, S., Han, B., Harari, R. E., Dias, R. D., Zenati, M. A., Salas, E., & Unhelkar, V. (2025, February 24). Socratic: Enhancing Human Teamwork via AI-enabled Coaching. arXiv.org. https://arxiv.org/abs/2502.17643

Our Implementation of Socratic AI Tutor





- Load actual course content from Semester 2 of COMP90059: Introduction to Programming
- Reduce additional workload, align AI uses with intended learning outcomes

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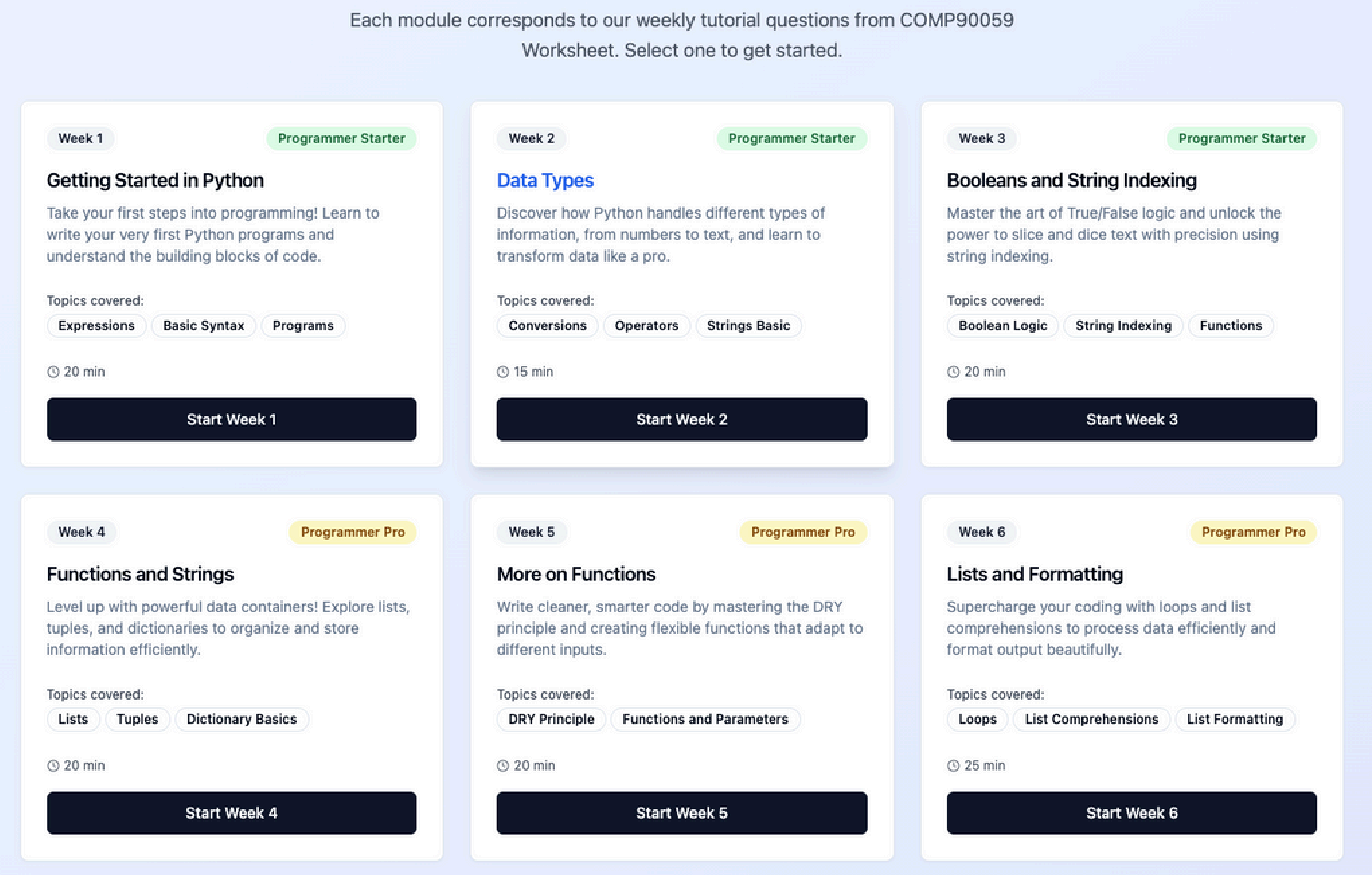

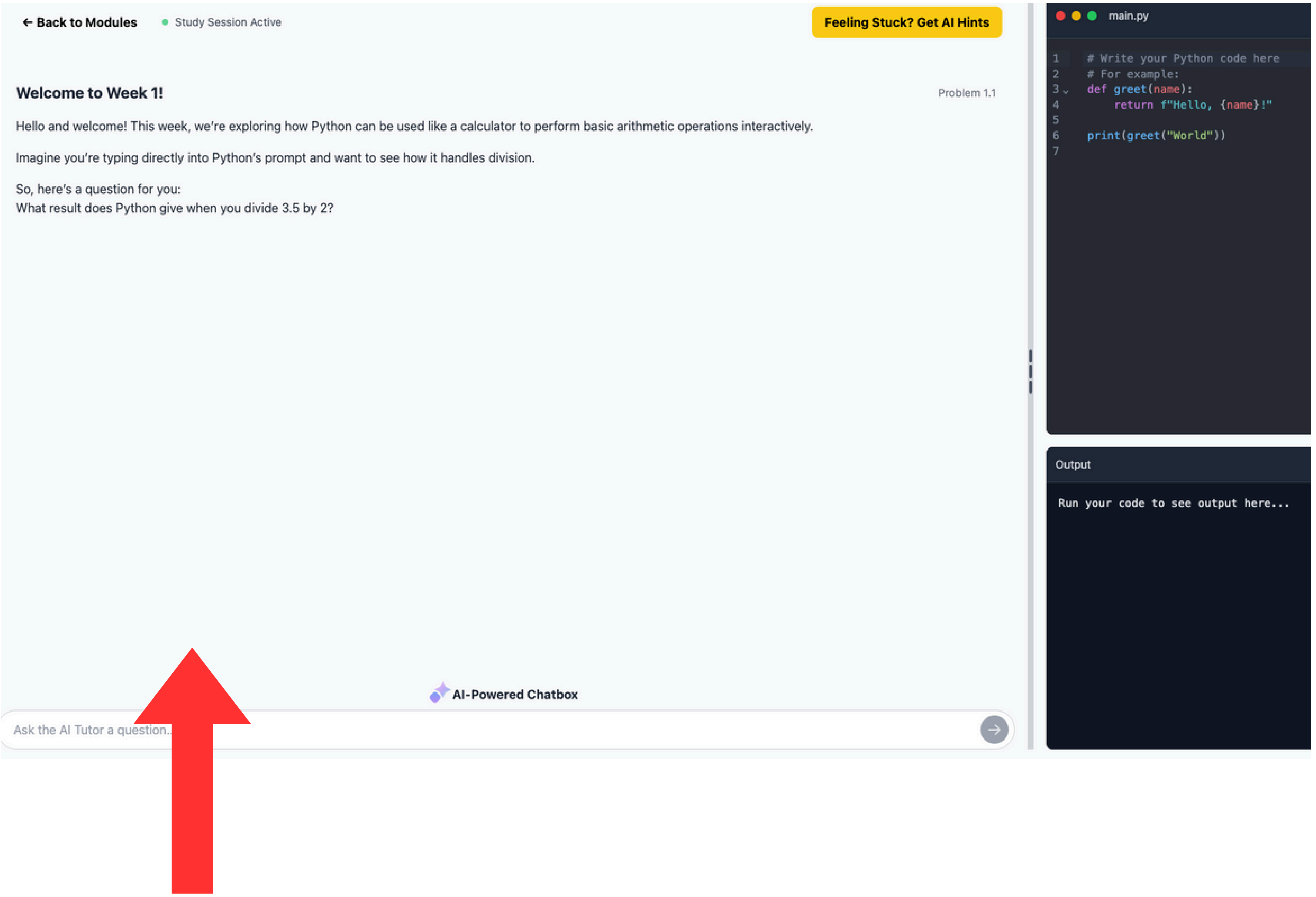

- **Main chat area, one problem at a time**
- AI progressive hints when they're stuck
- Built-in interpreter for code testing

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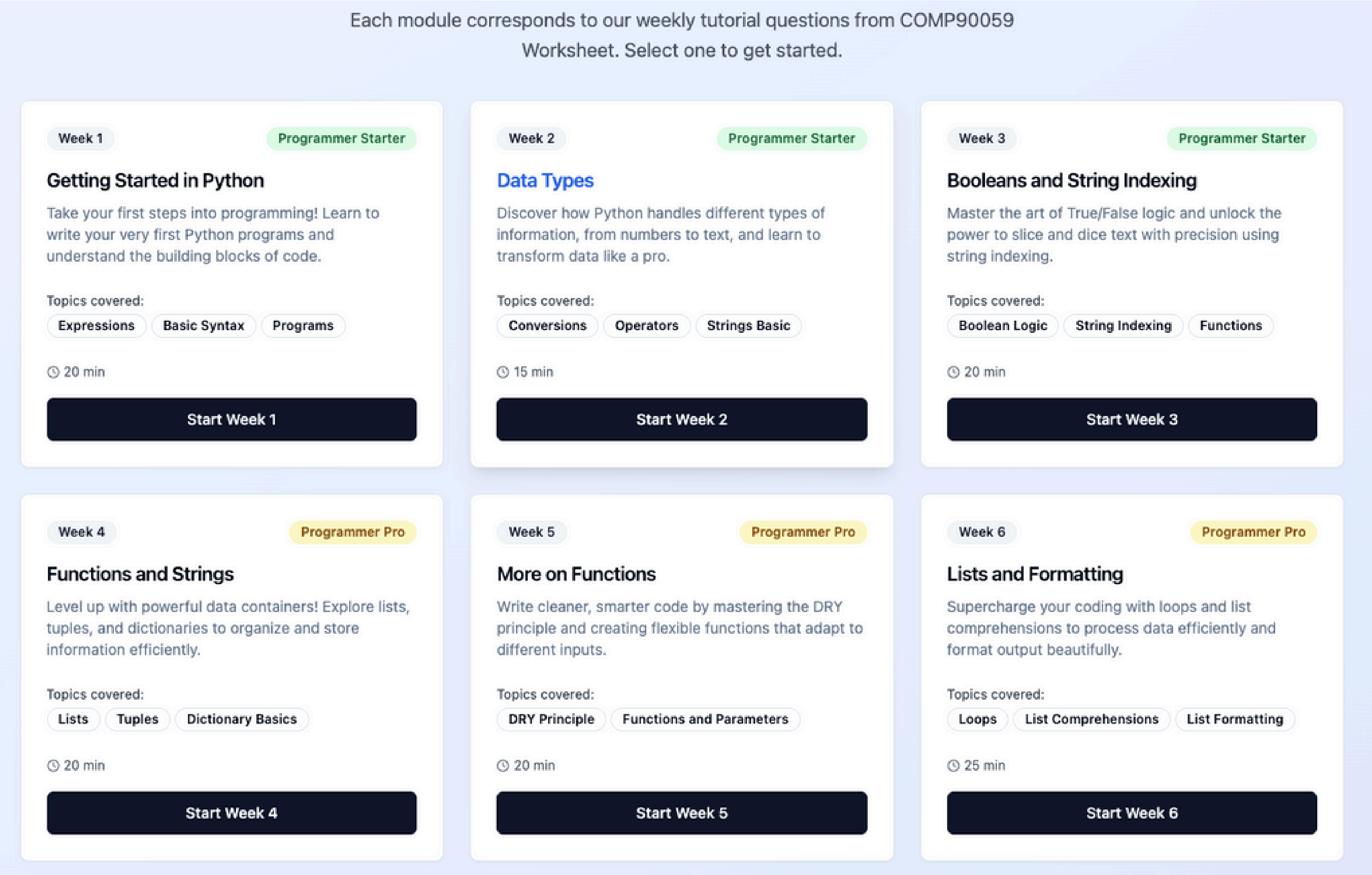

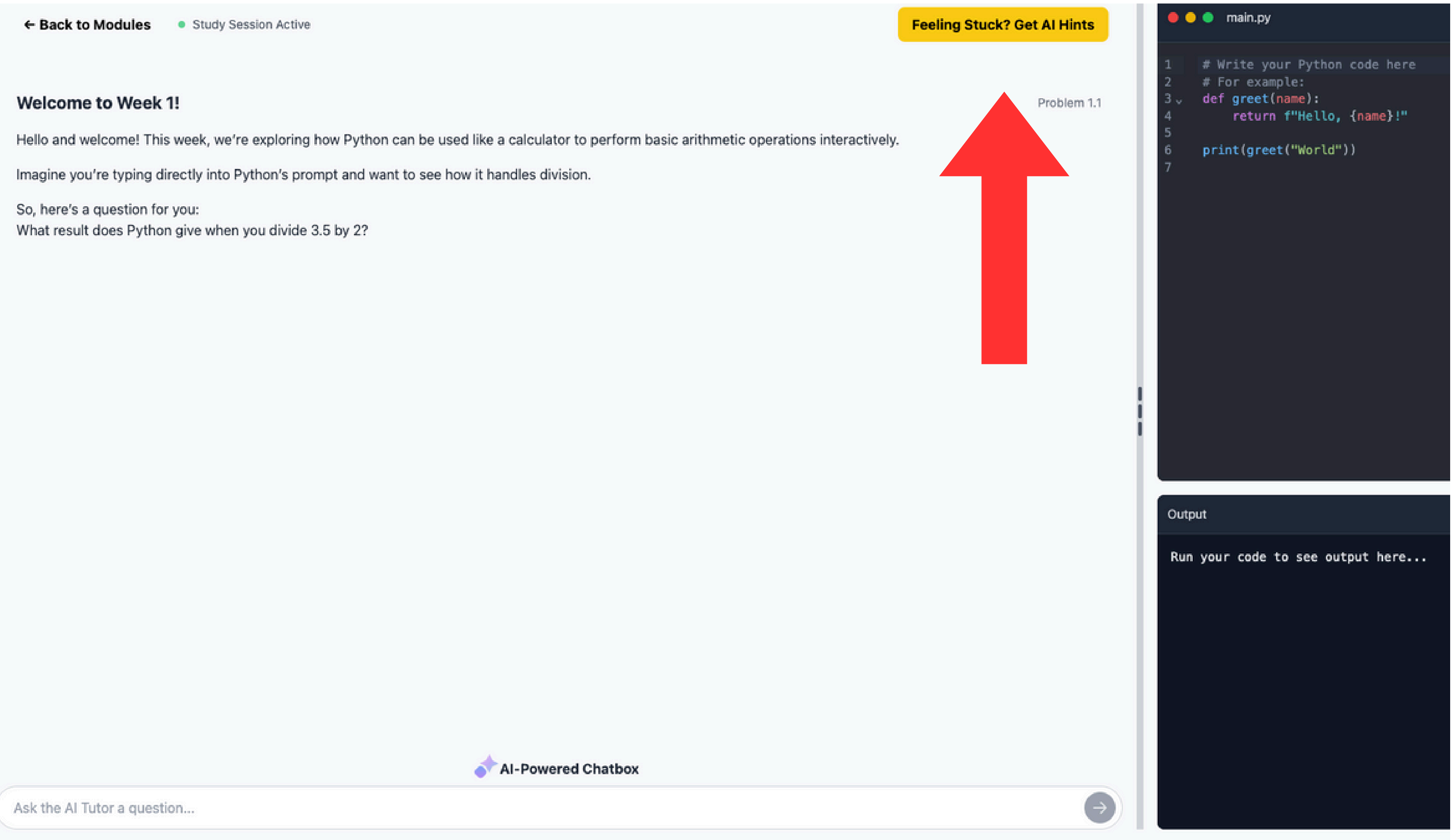

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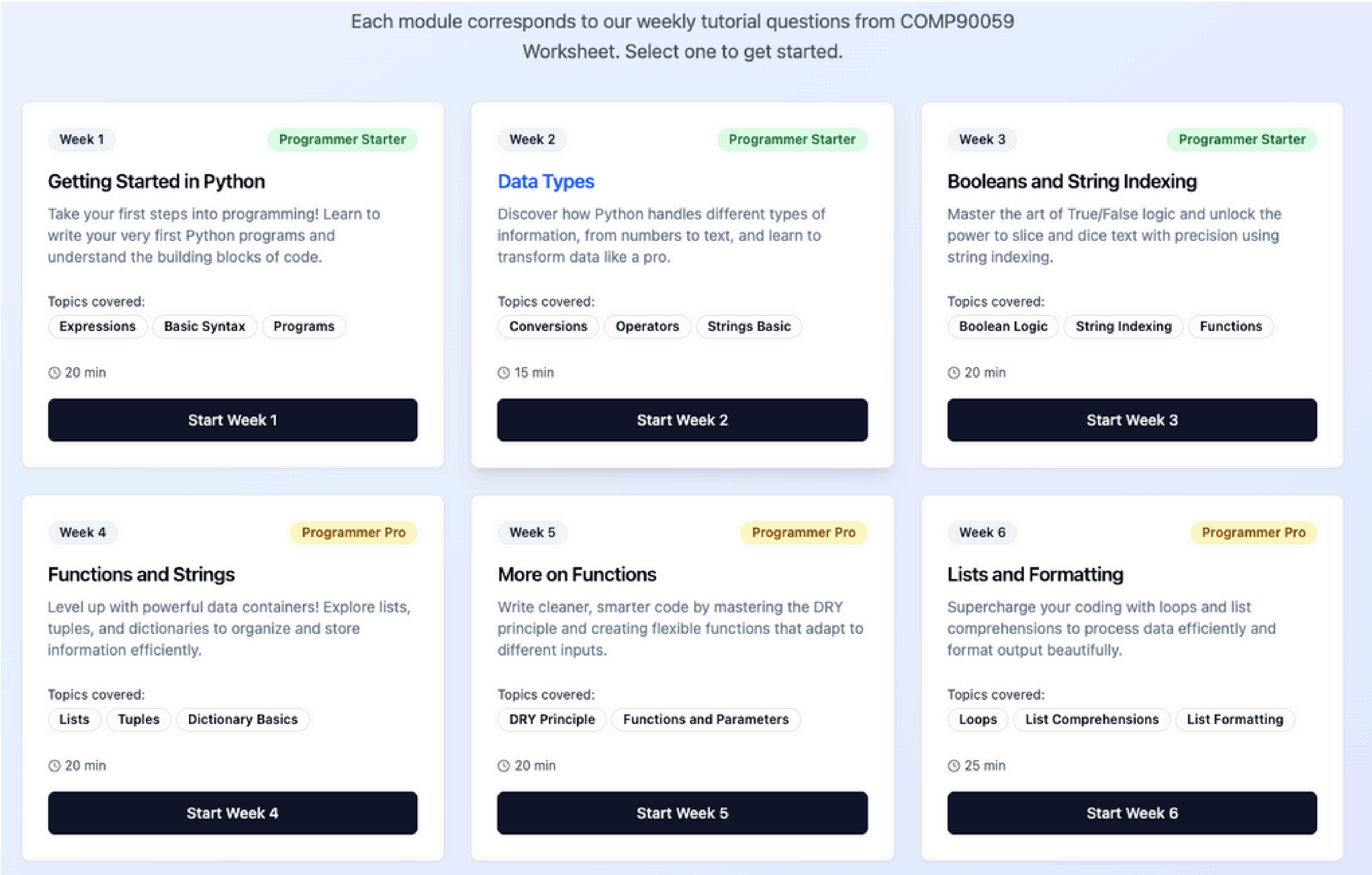

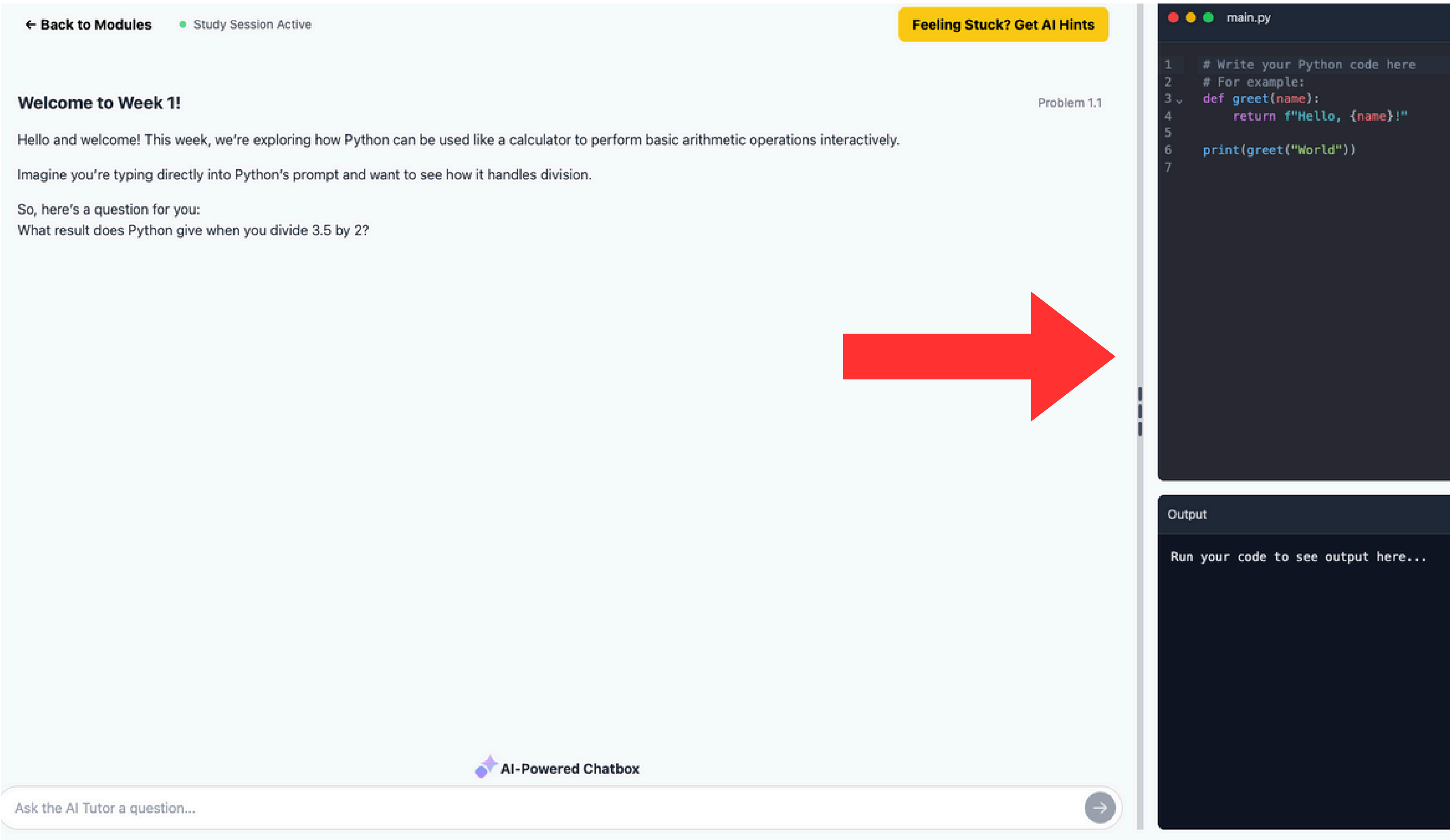

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The Result We Obtained

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Intended Analysis

Understanding the effect of AI usage towards the final course performance

Correlation between AI usage (i.e., frequency, duration, hints usage, etc.) and the exam score

Confounding factors control:

- pre-course survey & early assessment performance
- non-equivalent control group (i.e., students' performance from previous semester)

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Insufficient Uptake

Very low uptake

- n = 10 participants
- 26 conversation threads
- average of 3 utterances per session

Last text before drop-off:

- "What is the answer for this problem?" (n = 4)
- "Where to find more information on ....?" (n = 3)

Pivoting to our current study

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Literature and Reality Mismatch

In contrast to the enthusiasm shown in prior work, real-world implementation faced significant adoption issues:

Anecdotal evidence: COMP90059, SWEN90007, COMP100001, AILA & SPARKAI

Literature: Hanshaw & Sullivan, 2025 ; Russel et al., 2025

Well-known problems vs. less-known values/benefits

Issues of over-reliance, impacts on critical thinking, cognitive offloading extensively documented

Pedagogical benefits not well understood

Taking a bottom-up approach

Hanshaw, G., Sullivan, C. Exploring barriers to AI course assistant adoption: a mixed-methods study on student non-utilization. Discov Artif Intell 5, 178 (2025). https://doi.org/10.1007/s44163-025-00312-x

Jae-Eun Russell, Anna Marie Smith, Salim George, Jonah Pratt, Brian Fodale, Cassandra Monk, and Adam Brummett. 2025. Unlocking Insights: Investigating Student AI Tutor Interactions in a Large Introductory STEM Course. In Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK '25). Association for Computing Machinery, New York, NY, USA, 451–461. https://doi.org/10.1145/3706468.3706524

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Current Study Methodology

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Exploratory Sequential - Mixed Method Approach

Qualitative Phase

- Uncover use cases in programming education, their values, and where current AI still fails at meeting learners' needs

  SAMR Framework

- n >= 20, enough for theme saturation

Quantitative Phase

- Validating qualitative study at scale, produce generalisable insights
- Instruments (i.e., survey) informed from qualitative study
- n >= 300, enough for medium effect detection

Preliminary Findings

After 5 interviews conducted

Emerging Themes

Chatbot alone is insufficient

- Requires visual components
- Prefer redirection to more trustworthy documents

"Content Alignment" is a critical component of Personalisation

- Responses from AI should be tailored to course content
- Critical for preparing for assessments

AI cuts tutor requests while boosting outcomes

- Availability of AI reduces student's help seeking of tutors
- Yet, students feel more confident in their learning and material understanding

Future Planned Works

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Continue Conducting Interviews

- Requires at least 15 more participants
- Continuous coding for thematic analysis
- Findings write-up

Develop and Pilot Test Survey Instrument

- Develop survey based on findings
- Conduct pilot to test internal validity (i.e., Cronbach's alpha and Factor Analysis)

Thank you very much!

Any questions? Feedback?

Contact: ppit@student.unimelb.edu.au